



DEPARTMENT OF CHEMICAL ENGINEERING NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR

Subject: Bioresource Technology (CET-465)	Year & Semester: B.Tech. Chemical Engineering 4 th year & 8 th Semester		Total Course Credit: 4		
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			3	1	0
Evaluation Policy	Mid-Term (26 Marks)	Continuous Assessment (24 Marks = 8 Class Att. + 8 Assign.+8 Quiz)	Final-Term (50 Marks)		

Course Objective:

The aim of this course is to provide fundamental knowledge for bioenergy generation and product formation with the help of various conversion processes adequate to diverse bioresource characteristics.

Course Outcomes (COs):

CO1	Fundamental understanding of the bioresources and its applications for attainment of social objectives (energy, environment, product, sustainability).
CO2	Acquire knowledge with respect to the properties of the bioresources and the conversion technologies.
CO3	Exhibiting knowledge of the systems used for bioresource technology.
CO4	Understanding about analysis of data and their applications in design of the systems and development of the bioprocess.

Details of the Syllabus:

Unit-I	Bioresources- natural and anthropogenic; importance of bio-resources and their utilization. Natural bio-resources: agricultural, forestry and aquatic biomass. Biomass availability, production and food security, non- edible biomass characteristics. Anthropogenic bio-resources: Organic wastes-domestic and industrial; characteristics of municipal sewage / sludge and industrial sludges.
Unit-II	Conversion processes : biochemical, thermo-chemical and physico-chemical conversion processes. Biochemical processes : Microbial anaerobic and aerobic processes, enzymatic processes ; fermentation for alcohols and acids ; penicillin and other therapeutic products. Production of single cell protein (SCP) ; bio-pulping, biogasification. Thermo-chemical processes: pyrolysis (coke and pyro-oils), oxidation-

	combustion, gasification (downdraft, updraft and fixed bed gasification, fluidized bed and entrained bed gasification). Various methods of manufacture of activated carbons
Unit-III	Physico-chemical processes: Pretreatment, steam/acid/alkali hydrolysis, effect of temperature on hydrolysis.
Unit-IV	Special topics: biofuels , biomaterials, specialty chemicals (glycol, acetic acid and downstream chemicals), anhydrous alcohols-ethanol and butanol; biodiesel, bio-aviation turbine fuel (BATF).

Books Recommended

Text Books	1.	Shuler, M., Kargi, F., “ <i>Bioprocess Engineering, Basic Concep</i> ”, 2 nd Edn., Prentice Hall of India Pvt. Ltd. (2004).
	2.	Chakraverty, A., “ <i>Biotechnology and other Alternative Technologies</i> ”, Oxford and IBH Publishing Co. Pvt. Ltd. (1995).
	3.	Rao, M.G., Sittig, M., “ <i>Dryden’s Outlines of Chemical Technology- for the 21st Century</i> ”, East-West Press (1997).
	4.	Austin, G.T., “ <i>Shreve’s Chemical Process Industries</i> ”, McGraw-Hill Book Company (1984).
Reference Books	1.	Pandey, A., “ <i>Concise Encyclopaedia of Bioresource Technology</i> ”, CRC Press (2004).
	2.	Glaucia, M.S. et al. (eds), “ <i>Bioenergy & Sustainability: Bridging the Gaps</i> ”, SCOPE 72, Universidade de São Paulo, Brazil (2015).
	3.	Eckert & Trihn (eds), “ <i>Biotechnology for Biofuel Production and Optimization</i> ”, Elsevier (2016).
	4.	Cock, “ <i>Encyclopedia of Life Support Systems (EOLSS)</i> ”, UNESCO, (2011)
	5.	S. Van Loo, “ <i>Handbook of Biomass Combustion and Co-Firing</i> ”, Twente University Press, 2002.
	6.	Wang, W.C. et al., “ <i>Review of Biojet Fuel Conversion Technologies</i> ”, National Renewable Energy Laboratory (USDE), Technical Report, 2016.